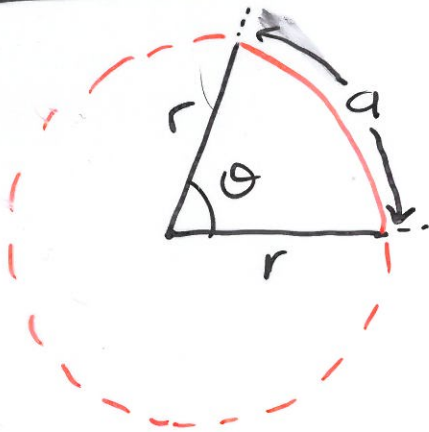


1.4 Radian measure and trigonometric functions

In calculus, we use radians to measure angles except when otherwise indicated.

Definition 1:



a : length of the arc
 r : radius of the circle
 θ : angle in radian

Circumference : $2\pi r$ ($\theta = 2\pi$ (~~360°~~))

$$\frac{\theta}{2\pi} = \frac{a}{2\pi r}$$

$\Rightarrow$

$$\theta = \frac{a}{r}$$

$$a = r\theta$$

$$\pi \text{ rad} = 180^\circ$$

Example 1:

(a) Convert the following angles in radians

(i) 90° ; (ii) 60° ; (iii) 50° .

(b) Convert in degrees the given angles in radians

(i) 5 rad ; (ii) $\frac{\pi}{9} \text{ rad}$

$$\begin{aligned} \therefore (a) (i) \quad \left. \begin{aligned} 2\pi &= 360^\circ \\ x &= 90^\circ \end{aligned} \right\} &\Rightarrow 2\pi \times 90^\circ = 360^\circ \cdot x \\ &\Rightarrow x = \frac{2\pi \cdot 90}{360} \\ &x = \frac{\pi}{2} \end{aligned}$$

$$(ii) \quad x = \frac{2\pi \cdot 60}{360} = \frac{2\pi}{3}$$

$$(iii) \quad x = \frac{2\pi \cdot 50}{360} = \frac{5\pi}{18}$$

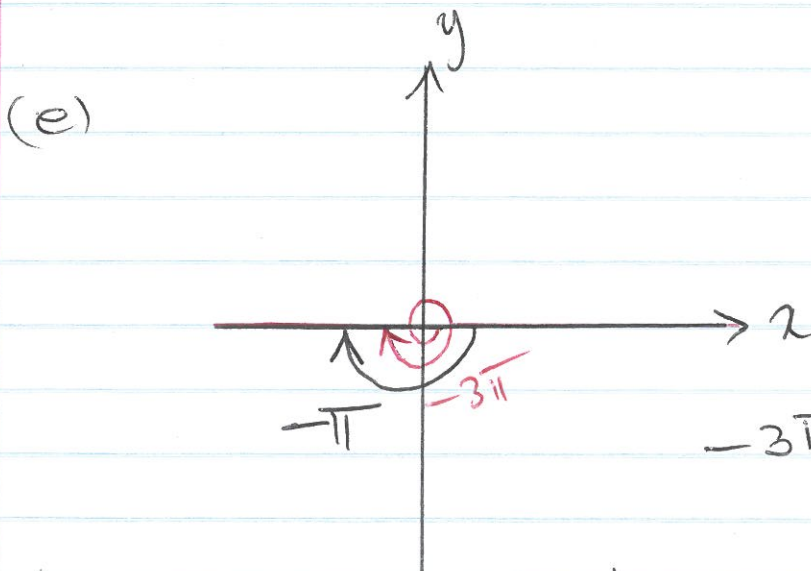
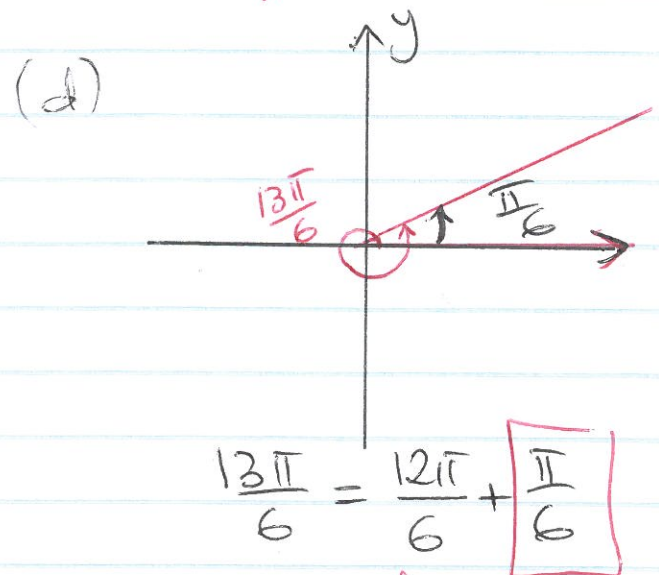
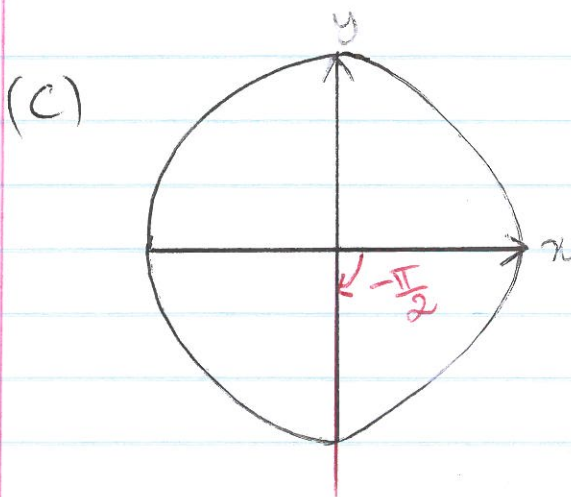
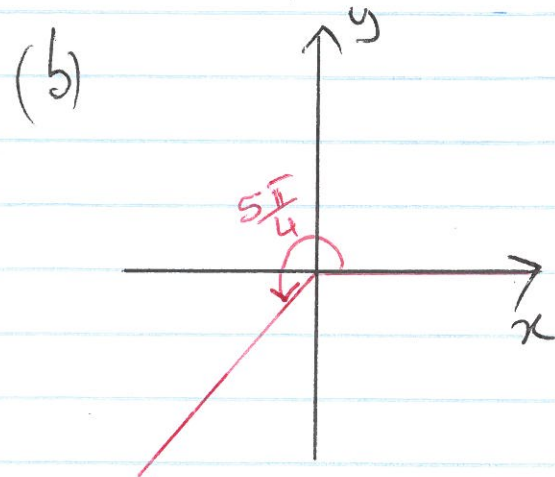
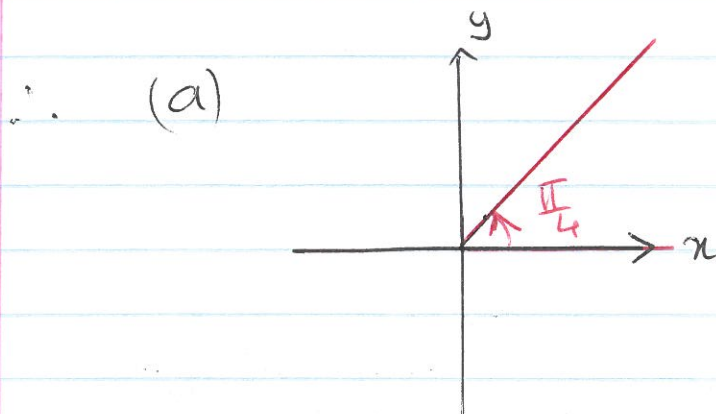
$$(b) (i) \quad \left. \begin{aligned} 360^\circ &= 2\pi \\ y &= 5 \end{aligned} \right\} &\Rightarrow y = \frac{5 \times 360}{2\pi} = \frac{5 \times 180^\circ}{\pi}$$

$$(ii) \quad \left. \begin{aligned} 360^\circ &= 2\pi \\ y &= \frac{\pi}{9} \end{aligned} \right\} &\Rightarrow y = \frac{\frac{\pi}{9} \times 360}{2\pi} = \frac{360}{18} = 20^\circ$$

Example 2 :

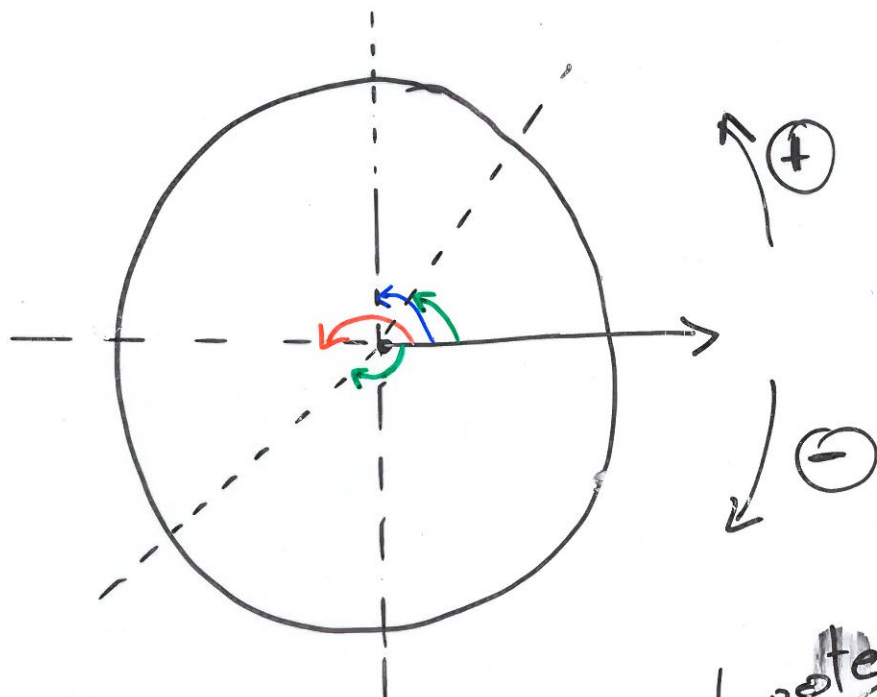
Sketch the following angles in a circle

(a) $\frac{\pi}{4}$; (b) $\frac{5\pi}{4}$; (c) $-\frac{\pi}{2}$; (d) $\frac{13\pi}{6}$; (e) -3π

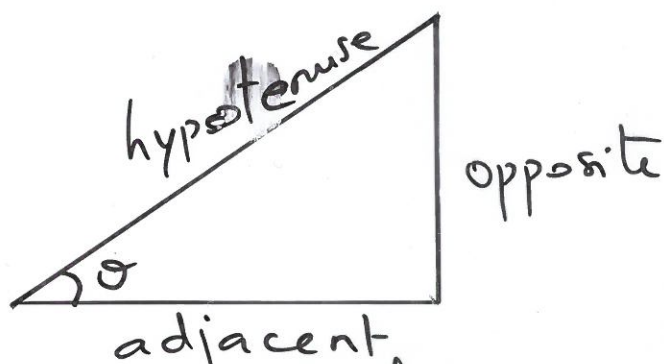


The standard position of θ_1 and θ_2 are the same if $\theta_2 = \theta_1 + 2k\pi$, $k \in \mathbb{Z}$

Degrees	0°	30°	45°	60°	90°	180°	360°
Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	2π



Definition 2 :



For an acute angle θ , the following trigonometric functions are defined

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

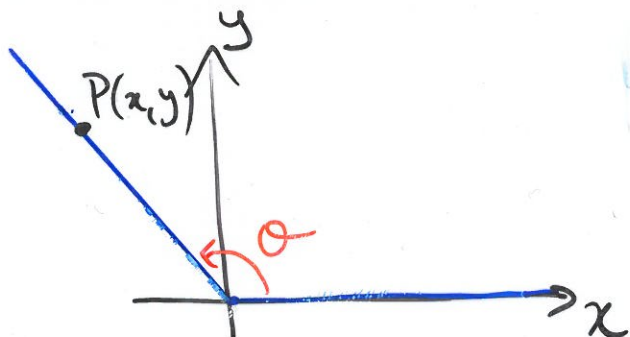
$$\csc \theta = \frac{1}{\sin \theta}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$



In general, with $r = |OP|$ the functions are defined as

$$\sin \theta = \frac{y}{r}$$

$$\cos \theta = \frac{x}{r}$$

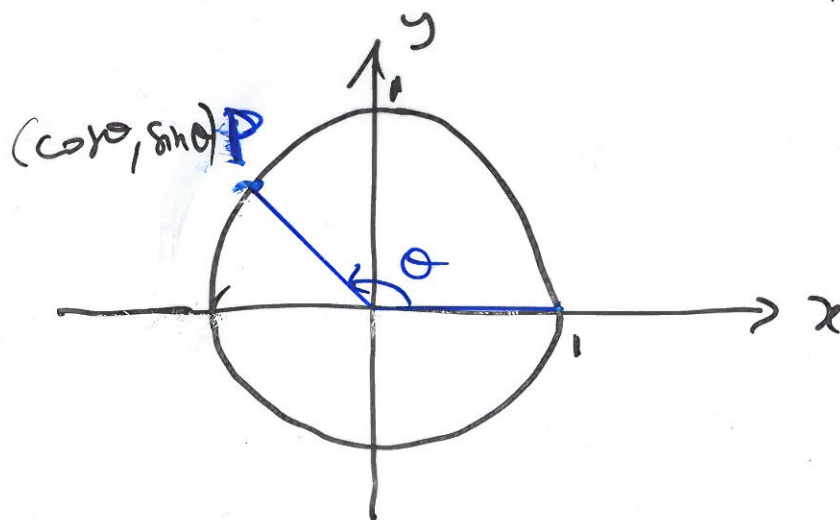
$$\tan \theta = \frac{y}{x}$$

$$\csc \theta = \frac{r}{y}$$

$$\sec \theta = \frac{r}{x}$$

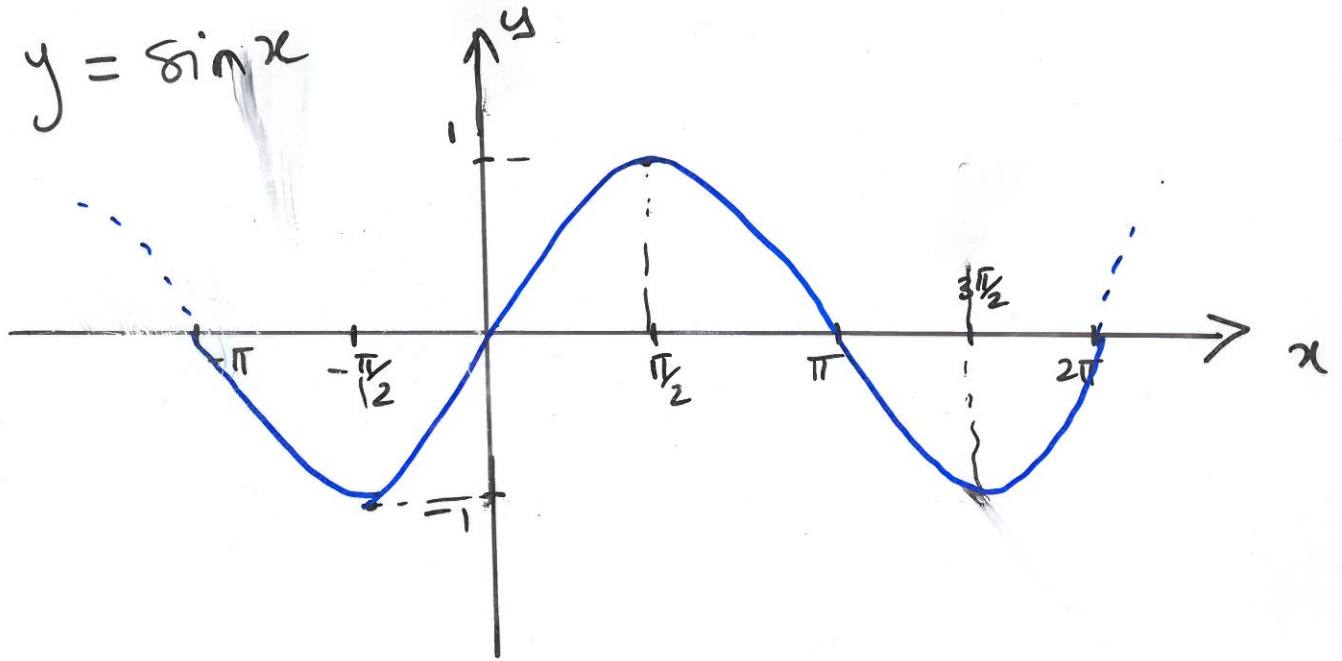
$$\cot \theta = \frac{x}{y}$$

On the unit circle ($r=1$), each point P has coordinates $(\cos \theta, \sin \theta)$

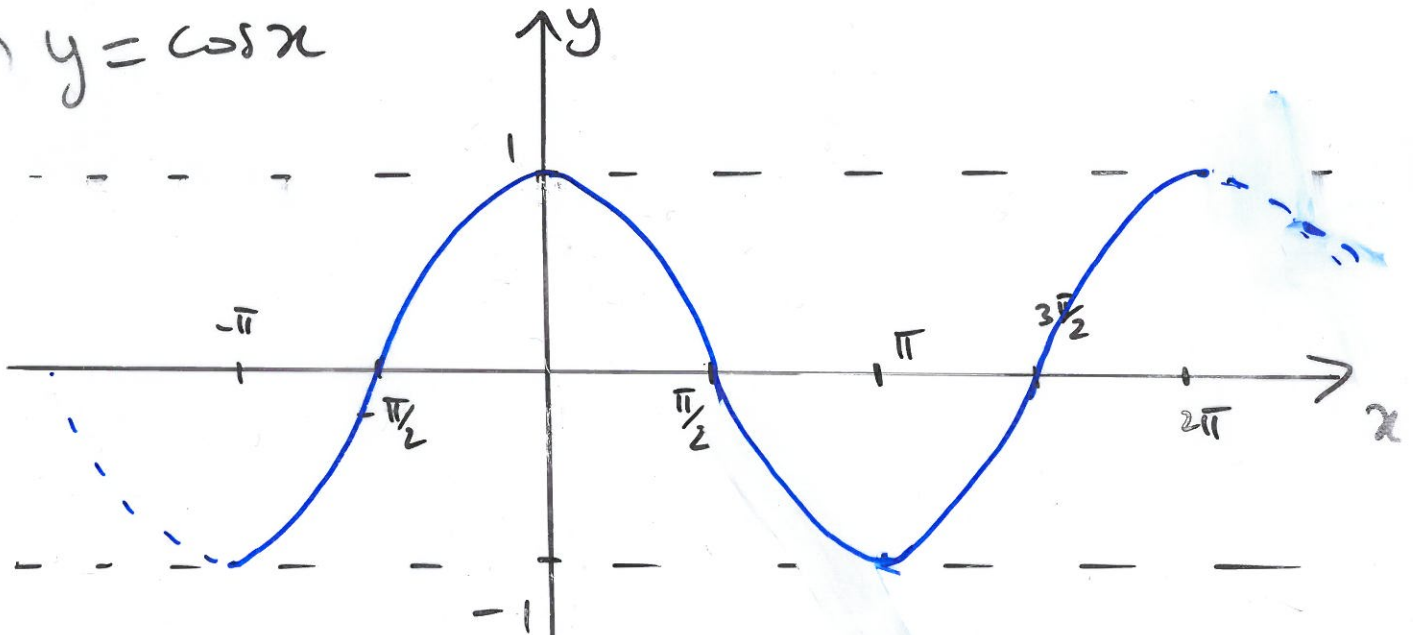


Graphs :

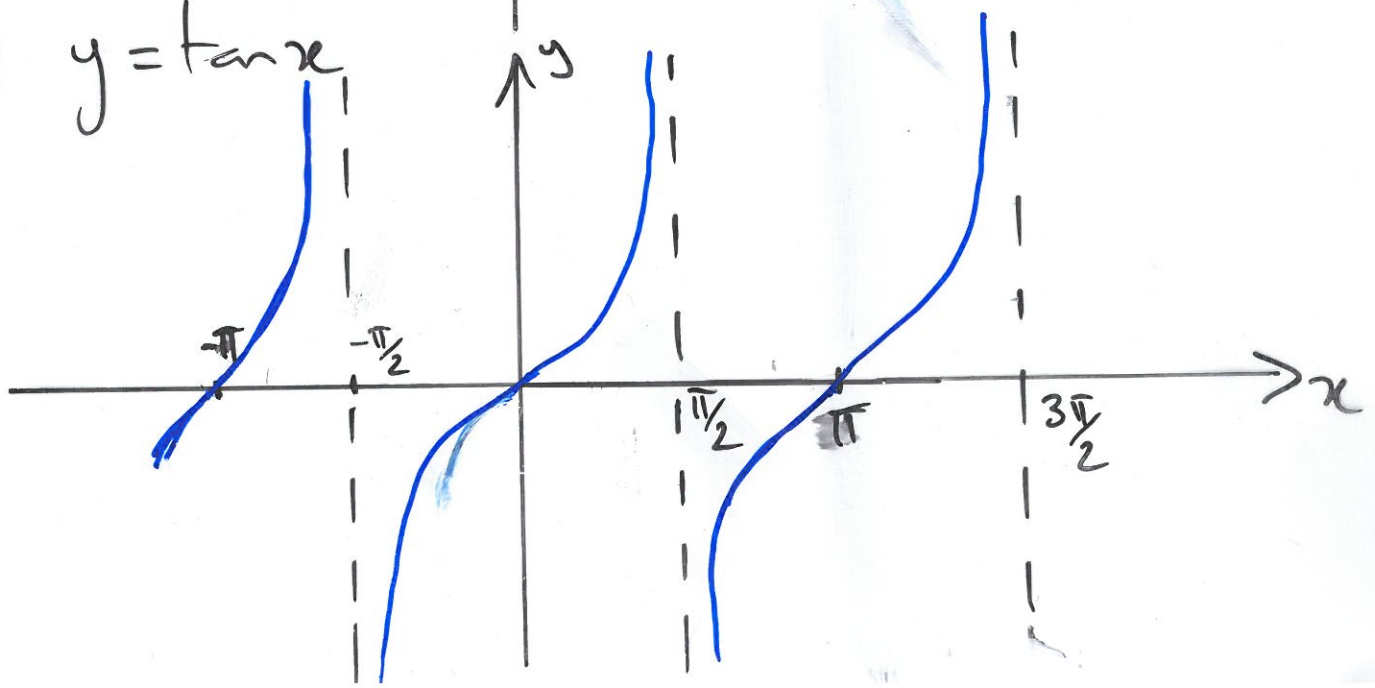
(i) $y = \sin x$



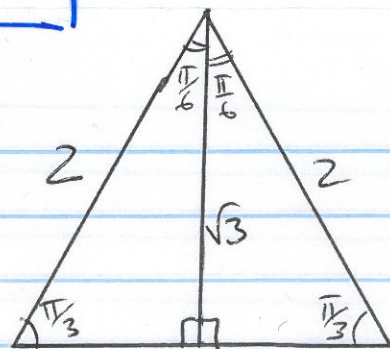
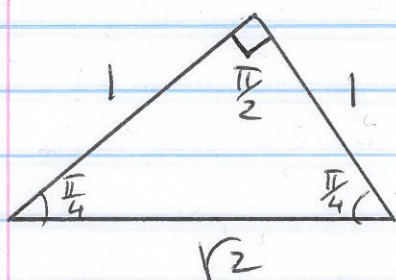
(ii) $y = \cos x$



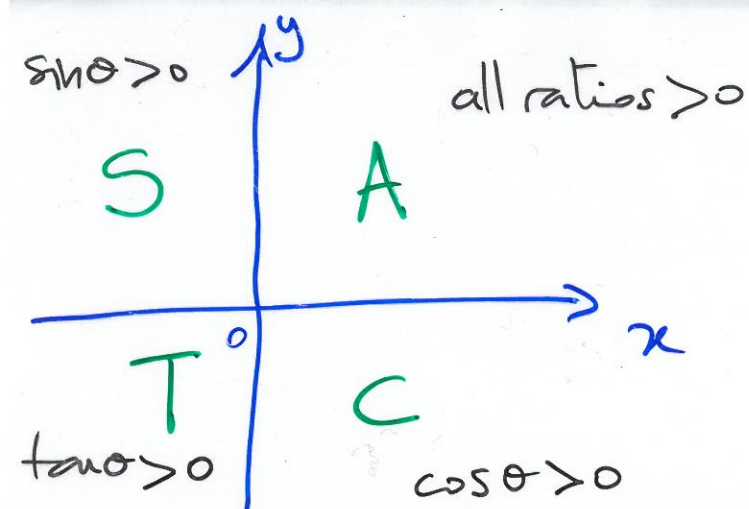
(iii) $y = \tan x$



radians	$\sin \theta$	$\cos \theta$	$\tan \theta$
0	0	1	0
$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$\frac{\pi}{2}$	1	0	—
π	0	-1	0
2π	0	1	0

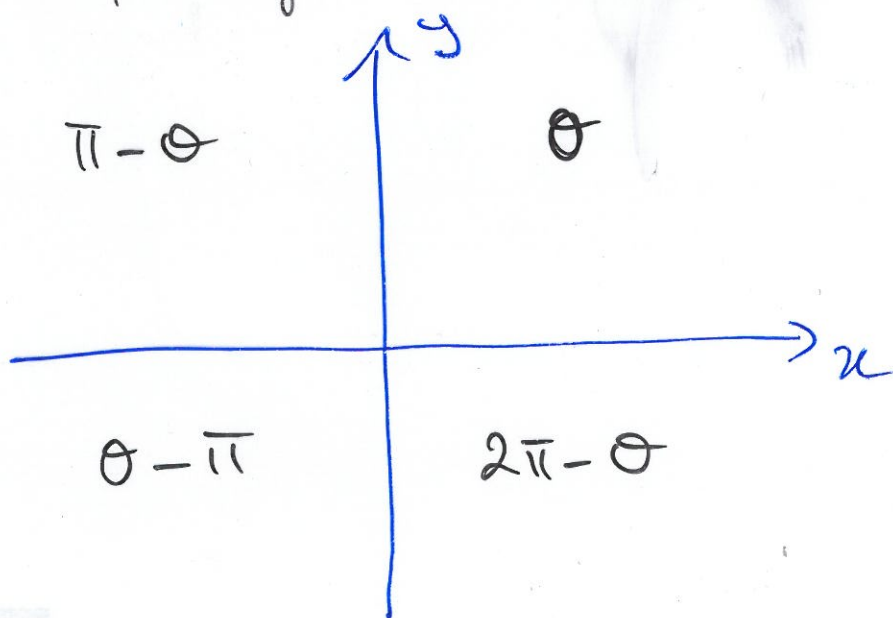


The above triangles can be used to remember the values of $\sin \theta$, $\cos \theta$ and $\tan \theta$ at the particular angles $\theta \in \left\{ \frac{\pi}{3}, \frac{\pi}{4}, \frac{\pi}{6} \right\}$



Definition 4: The reference angle is the smallest angle that the terminal side of a given angle makes with the x-axis.

Hence θ is a reference angle if and only if $\theta \in [0, \frac{\pi}{2}]$



Example 3:

(1) If $\tan \theta = -2$, $\frac{\pi}{2} < \theta < \pi$, find the values of the other 5 trigonometric ratios

(2) Solve for x if (i) $2\sin x - 1 = 0$, $x \in [0, 2\pi]$

(ii) $\cos x = -\frac{\sqrt{3}}{2}$, $x \in \mathbb{R}$

(iii) $\tan x = 1$, $x \in (0, 2\pi)$

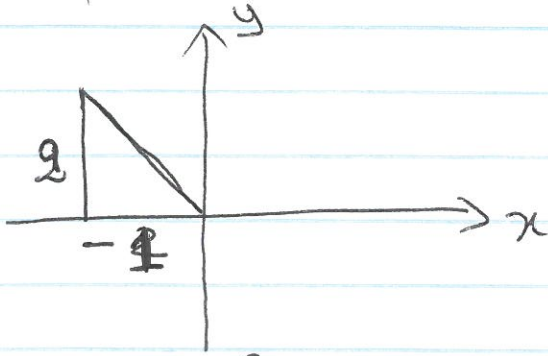
(3) Find the trigonometric ratios

if (i) $\theta = \frac{5\pi}{4}$; (ii) $\theta = \frac{20\pi}{2}$

(iii) $\theta = -\frac{2\pi}{3}$

Example 3 :

(1)



$$r = \sqrt{1^2 + (-2)^2} = \sqrt{5}$$

$$\text{Hence } \cos \theta = \frac{-1}{\sqrt{5}}$$

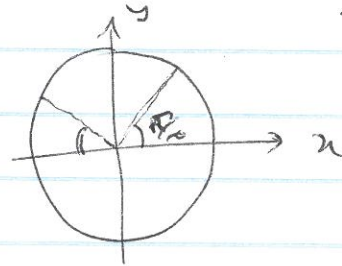
$$\sin \theta = \frac{2}{\sqrt{5}} ; \sec \theta = -\sqrt{5} ; \cot \theta = -\frac{1}{2}$$
$$\csc \theta = \frac{\sqrt{5}}{2}$$

(2) (i) $2 \sin x - 1 = 0 \Rightarrow \sin x = \frac{1}{2}$

$$\Rightarrow \text{RA : } x = \frac{\pi}{6}$$

$$\Rightarrow x = \frac{\pi}{6} \text{ or } x = \frac{5\pi}{6}$$

$$\Rightarrow x \in \left\{ \frac{\pi}{6}, \frac{5\pi}{6} \right\}$$



(ii) $\cos x = -\frac{\sqrt{3}}{2} \Rightarrow \text{RA : } x = \frac{\pi}{6}$

$$x = \pi - \frac{\pi}{6} = \frac{5\pi}{6} \text{ or } x = \pi + \frac{\pi}{6} = \frac{7\pi}{6}$$

$$\text{since } x \in \mathbb{R}, x = \frac{5\pi}{6} + 2k\pi, k \in \mathbb{Z}$$

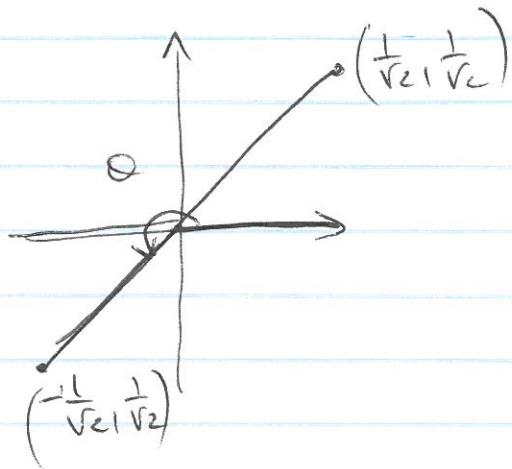
$$\text{or } x = \frac{7\pi}{6} + 2k\pi, k \in \mathbb{Z}.$$

(iii) $\tan x = 1 \Rightarrow \text{RA : } x = \frac{\pi}{4}$

$$\Rightarrow x = \frac{\pi}{4} \text{ or } x = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$$

(3) (i) $\theta = \frac{5\pi}{4}$, $RA = \frac{\pi}{4}$ ($\pi - \frac{\pi}{4}$)

$$\sin\left(\frac{5\pi}{4}\right) = -\frac{1}{\sqrt{2}} ; \cos\left(\frac{5\pi}{4}\right) = -\frac{1}{\sqrt{2}}$$



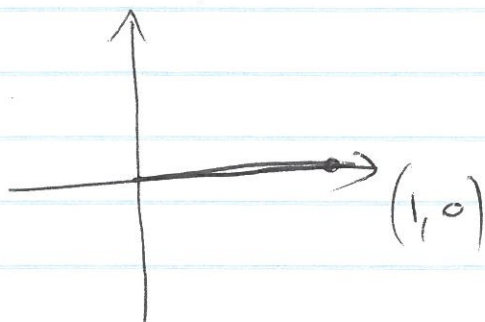
$$\tan\left(\frac{5\pi}{4}\right) = 1$$

$$\sec\left(\frac{5\pi}{4}\right) = -\sqrt{2}$$

$$\csc\left(\frac{5\pi}{4}\right) = -\sqrt{2}$$

$$\cot\left(\frac{5\pi}{4}\right) = 1$$

(ii) $\theta = \frac{20\pi}{2} = 10\pi$, $RA = 0$



$$\sin(10\pi) = 0$$

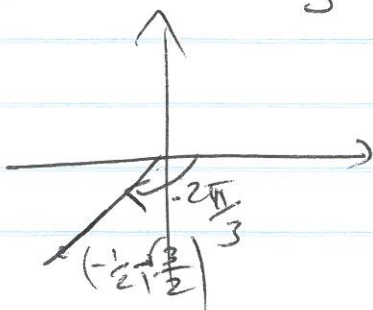
$$\cos(10\pi) = 1$$

$$\tan(10\pi) = 0$$

$$\sec(10\pi) = 1$$

$\cot$ and $\csc$ are undefined at 10π .

(iii) $\theta = -\frac{2\pi}{3}$, $RA = \frac{\pi}{3}$



$$\sin\left(-\frac{2\pi}{3}\right) = -\frac{\sqrt{3}}{2} ; \cos\left(-\frac{2\pi}{3}\right) = -\frac{1}{2}$$

$$\tan\left(-\frac{2\pi}{3}\right) = \sqrt{3} ; \sec\left(-\frac{2\pi}{3}\right) = -2$$

$$\csc\left(-\frac{2\pi}{3}\right) = -\frac{2}{\sqrt{3}} ; \cot\left(-\frac{2\pi}{3}\right) = \frac{1}{\sqrt{3}}$$

Trigonometric identities :

$$\cos^2 x + \sin^2 x = 1$$

$$\sin(x + 2\pi) = \sin x \quad ; \quad \cos(x + 2\pi) = \cos x$$

$$\sin(-x) = -\sin x \quad ; \quad \cos(-x) = \cos x$$

$$\sin(2x) = 2 \sin x \cdot \cos x$$

$$\cos^2 x = \frac{1 + \cos 2x}{2} \quad ; \quad \sin^2 x = \frac{1 - \cos 2x}{2}$$

$$\tan^2 x + 1 = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \cdot \tan y}$$

Example 4: Find the values of $x \in [0, 2\pi]$ such that

1. $\sin x \leq \frac{1}{2}$

2. $\sin x > \cos x$

3. $|2\sin x| < 1$

4. $\tan x < 1$

5. $3\cot^2 x = 1$

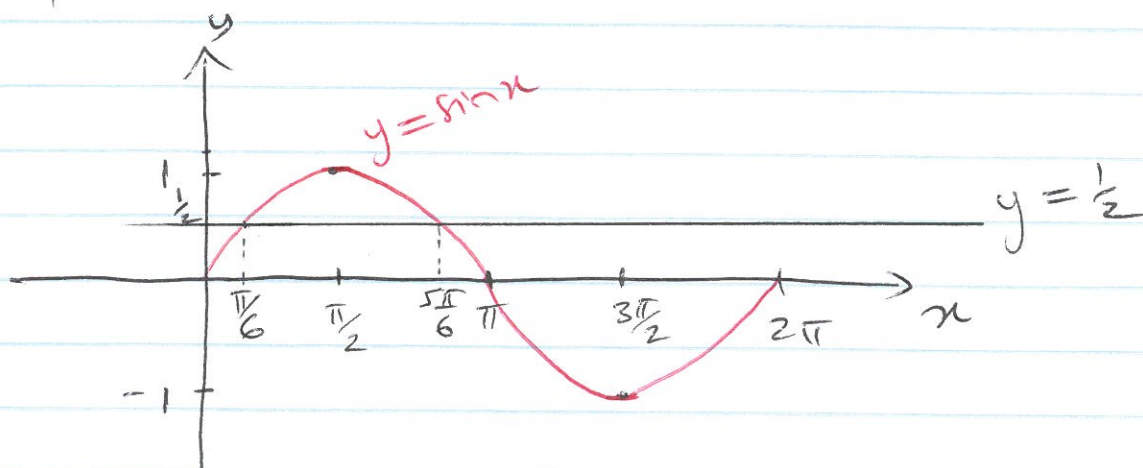
6. $2 + \cos 2x = 3\cos x$

7. $\csc x = \cot x$

8. $\sin 2x = \cos x$

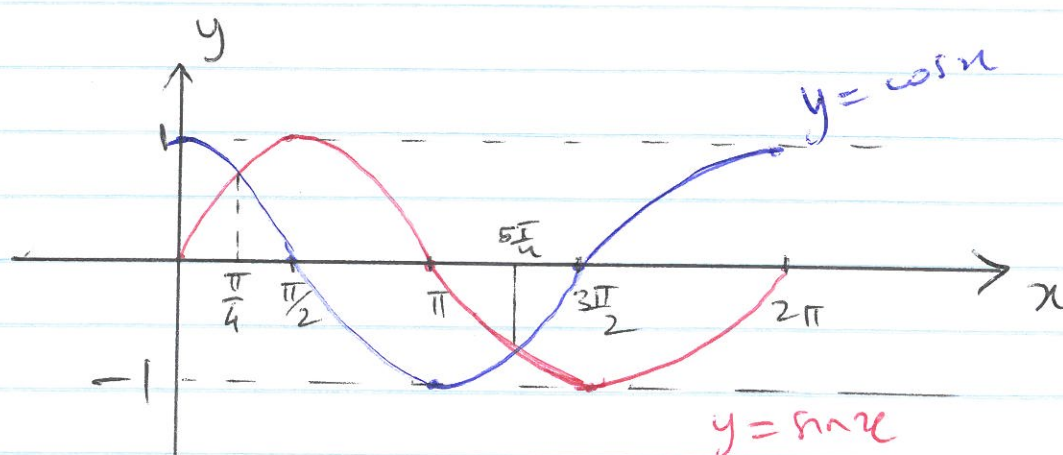
Example 4

1.



$$x \in \left[0, \frac{\pi}{6}\right] \cup \left[\frac{5\pi}{6}, 2\pi\right]$$

2.



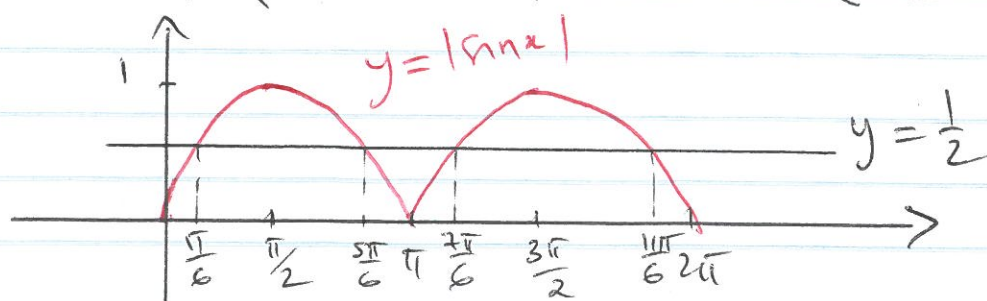
$$x \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$$

$$3. |2 \sin x| < 1 \Rightarrow -1 < 2 \sin x < 1$$

$$-\frac{1}{2} < \sin x < \frac{1}{2}$$

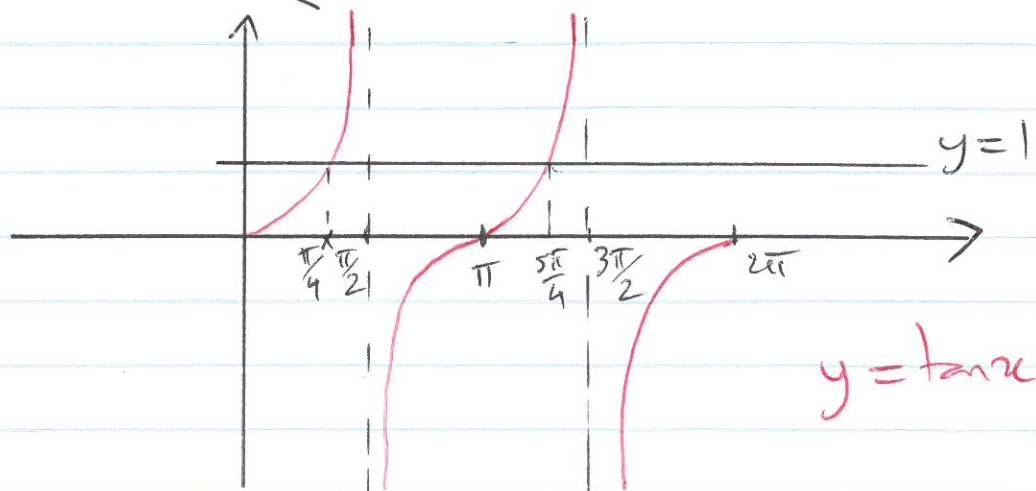
or

$$|2 \sin x| < 1 \Rightarrow |\sin x| < \frac{1}{2}$$



$$x \in [0, \frac{\pi}{6}) \cup (\frac{5\pi}{6}, \frac{7\pi}{6}) \cup (\frac{11\pi}{6}, 2\pi]$$

$$4. \tan x < 1$$



$$x \in [0, \frac{\pi}{4}) \cup (\frac{\pi}{2}, \frac{5\pi}{4}) \cup (\frac{3\pi}{2}, 2\pi]$$

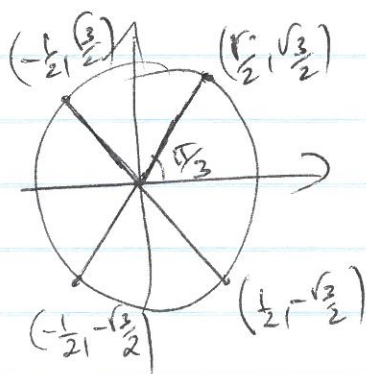
$$5. 3 \cot^2 x = 1 \Rightarrow \cot^2 x = \frac{1}{3}$$

$$\Rightarrow \cot x = \pm \frac{1}{\sqrt{3}} \quad (\tan x = \pm \sqrt{3})$$

$$\text{RA : } x = \frac{\pi}{3}$$

$$\Rightarrow x = \frac{\pi}{3}, \frac{4\pi}{3} \text{ or } x = \frac{2\pi}{3}, \frac{5\pi}{3}$$

$$\Rightarrow x \in \left\{ \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3} \right\}$$



$$6. \quad 2 + \cos 2x = 3 \cos x$$

$$2 + 2 \cos^2 x - 1 = 3 \cos x$$

$$2 \cos^2 x - 3 \cos x + 1 = 0$$

$$(2 \cos x - 1)(\cos x - 1) = 0$$

$$\cos x = \frac{1}{2} \quad \text{or} \quad \cos x = 1$$

$$x \in \left\{ \frac{\pi}{3}, \frac{5\pi}{3} \right\} \quad \text{or} \quad x \in \{0, 2\pi\}$$

$$x \in \left\{ 0, \frac{\pi}{3}, \frac{5\pi}{3}, 2\pi \right\}$$

$$7. \quad \csc x = \cot x$$

$$\Rightarrow \frac{1}{\sin x} = \frac{\cos x}{\sin x}$$

$$\Rightarrow \sin x = \cos x \cdot \sin x$$

$$\Rightarrow \sin x (1 - \cos x) = 0$$

$$\Rightarrow \sin x = 0 \quad \text{or} \quad \cos x = 1$$

$$\Rightarrow x = 0, \pi, 2\pi \quad \text{or} \quad x = 0, 2\pi$$

$$\Rightarrow x \in \{0, \pi, 2\pi\}$$

$$8. \sin 2x = \cos x$$

$$\Rightarrow 2 \sin x \cos x = \cos x$$

$$\Rightarrow \sin x = \frac{1}{2} \Rightarrow x \in \left\{ \frac{\pi}{6}, \frac{5\pi}{6} \right\}$$